



# Prosthetic Foot Prototype

 **Make3D Company Limited**

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## Summary

This is our new prosthetic foot prototype. It has not been properly tested yet.

[Healthcare](#) > [Medical Tools](#)

Tags: [printableprosthetics](#) [prosthetic](#)

This is a new 3D printable prosthetic foot with the connector to pylon used on conventional prosthetics. It has the ankle joint and flexible fingers to achieve better walking dynamics.

The design was made in Autodesk Fusion 360 and was printed in several iterations before we have achieved the right fit, function and strength.

This design has not been tested properly yet. Use on your own risk.

### Print Settings:

#### TopJoint, Inserts,InnerFoot

- Perimeters: 5, Infill: 100%, Layer height: 0.25mm

### **RubberCube:**

- Filament: Filaflex40, Infill: 20-40% (depending on the weight of the person and required rigidity), Layer height: 0.25mm

### **FootRubber:**

- Filament: Filaflex40, Infill: 20-40% (depending on the weight of the person and required rigidity), Layer height: 0.25mm

### **Additional material:**

1. 3 bolts, nuts and washers for attaching the joint to the foot
2. 1 metal cylinder diameter 10mm, length 20mm
3. Strong spring - depending on the weight of the person, sometimes we use two springs inside each other or a 3D printed rubber block for higher rigidity.

### **ASSEMBLING:**

1. Glue the side joints on the top foot, and use size 10 bolt and nut to tighten them
2. Insert the Flex Cube in the top foot
3. Insert the inside foot in the rubber foot. Use glue to attach the parts together firmly. It might be very tricky to insert the parts together. When using the tools pay attention not to damage the parts or hurt yourself.
4. Pour glue in the hole found on the top part part of the foot and Insert a metal cylinder (stainless steel is preferred) in the hole.
5. Insert the spring between the foot and top joint in between the pins and add washers between the spring and 3D printed parts (we recommend to glue the washers).
6. Insert the joint on the foot.
7. Attach a washer between and after the side joints , and insert the bolt.
8. Tighten a bolt and nut slightly. Do not over tighten the nut.
9. Attach the conventional pylon joint on the top and tighten it with the bolt. Dont forget to insert nut and washer on the side of the top part.

## **Model files**



**insert.stl**

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**insert.stl**

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**rubbercube.stl**

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**topjoint.stl**

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**innerfoot.stl**

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**footrubber.stl**

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